Companion access after black-hole recoil: dynamical benchmarks and sparse binary populations

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The origin and growth of merging black holes are central questions in gravitational-wave astronomy. Dense star clusters can support repeated mergers, but gravitational-wave recoil can displace a remnant from regions where encounters occur without ejecting it. Continued growth therefore depends on companion access as well as retention. We examine how accurately simplified descriptions preserve that access using orbital benchmarks, velocity-dependent passage calculations, and audits of released cluster simulations. In a static Plummer potential, replacing recoil motion by a delay followed by permanent central residence misses early density exposure and overestimates it at late times. In an imposed binary population, density exposure alone also overestimates normalized passage counts for the selected larger kicks because it omits relative-velocity weighting. A frozen cluster snapshot shows that matching total mass and half-mass radius does not preserve ballistic return times. A wider census finds small simultaneous binary-black-hole populations throughout the inspected late-time states, while the available summaries lack the joint properties needed for encounter calibration. Together, these results identify limitations of reduced access prescriptions and motivate a testable hypothesis: temporal overlap between remnant motion and evolving target availability improves passage predictions over stationary population descriptions conditional on remnant-producing mergers. Establishing that improvement requires merger-linked histories and resolved trajectories; companion acquisition and repeated-merger rates remain uncalibrated.

I. INTRODUCTION

Repeated mergers in dense star clusters offer a route to black-hole growth and a possible origin for massive gravitational-wave sources [1, 2]. Asymmetric gravitational-wave emission imparts recoil to the remnant, which can remain bound while leaving the dense part of its host. Another merger requires the remnant to reach suitable companions, acquire or enter a binary, and undergo subsequent evolution. Relating this sequence to gravitational-wave source populations requires knowing which environmental information survives a reduced dynamical description. We focus on one link: *how accurately can remnant motion and an evolving binary population be reduced to a prediction of companion access?* Here access means a declared close passage to an existing binary black hole (BBH); other acquisition channels and the subsequent encounter outcome require separate treatment.

Published work already treats recoil displacement and frictional recentering [1], host evolution and pairing [3], and successive mergers with explicit few-body encounters [2]. Mass segregation and the coupling of black-hole subsystems to their hosts are established [4]. Small coexisting BBH populations and their formation, destruction, and ejection have also been reported [5]. The opportunity addressed here is to test the adequacy of reduced accessibility prescriptions against these physical ingredients, rather than to introduce them as new mechanisms.

We organize the investigation by the assumption being tested. A ballistic-return benchmark in a Plummer potential [6] establishes the orbit; residence and density-exposure comparisons then test whether a central-return prescription preserves its environmental sampling. A specified tracer-binary bath isolates the effect of velocity weighting on close-passage counts. An evolved Cluster Monte Carlo (CMC) snapshot tests the assumed target population and potential shape [7, 8]. Finally, an age-selected audit of available late-time summaries tests whether the snapshot's small BBH population is confined to that one state. Each comparison changes the information being supplied, rather than merely enlarging the same kick grid.

The results motivate the hypothesis that temporal overlap between remnant motion and changing target availability adds predictive information beyond a return time and stationary target description. Section IX B defines the corresponding conditional quantity and a comparison that could reject the hypothesis. The present work supplies its motivation and approximation tests; it does not measure that overlap in live clusters. In particular, the passage calculation uses an imposed bath, the frozen-potential comparison uses a different host and physical kicks, and the population census is not conditioned on mergers. Their discrepancies cannot be combined into a calibrated rate correction.

Ballistic center crossing is an orbital event; frictional recentering requires dissipative evolution omitted here. Within the fixed one-parameter Plummer family, an exact return time and the governing equations can identify the orbit; the error lies in replacing it by an assumed residence history. No observed gravitational-wave signals, radiation waveforms, or merger outcomes are analyzed. The long-term physical objective is to establish which assembly-history information an accessibility model can carry into predictions of repeated mergers; this paper tests the inputs needed for that step.

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II. MODEL AND NUMERICAL METHOD

With host mass M, scale radius b, and gravitational constant G, define

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + b^2}}, \tag{1}$$

$$t_d = \sqrt{\frac{b^3}{GM}}, \qquad \eta = \frac{v_k}{\sqrt{2GM/b}}. \tag{2}$$

Let x be signed displacement in units of b and let time be measured in t_d. The initial-value problem is

$$\ddot{x} = -\frac{x}{(1 + x^2)^{3/2}}, \tag{3}$$

$$x(0) = 0, \qquad \dot{x}(0) = \sqrt{2}\eta, \tag{4}$$

with conserved dimensionless specific energy $E = \eta^2 - 1$. We restrict attention to $0 \le \eta < 1$. For a nonzero kick, P is the interval between successive center crossings. Signed position has period $2P$; radius and density have period P. The exactly stationary case has no return event and is treated separately.

A. Ballistic return benchmark

Energy conservation yields the apocenter

$$x_a = \frac{\eta\sqrt{2 - \eta^2}}{1 - \eta^2} \tag{5}$$

and return time

$$P = 2\int_0^{x_a} \frac{dx}{\sqrt{2[(1 + x^2)^{-1/2} - (1 - \eta^2)]}}. \tag{6}$$

Set $x = x_a \sin\theta$, $s = \sqrt{1 + x_a^2 \sin^2\theta}$, and $s_a = (1 - \eta^2)^{-1}$. The endpoint singularity is removed:

$$\frac{dt}{d\theta} = \sqrt{\frac{s s_a (s + s_a)}{2}}, \tag{7}$$

$$P = 2\int_0^{\pi/2} \frac{dt}{d\theta}\, d\theta. \tag{8}$$

B. Residence and density-exposure comparison

For horizon H and operational sphere radius cb, define

$$C(H, c) = \int_0^H \mathbf{1}_{|x(t)|<c}\, dt, \tag{9}$$

$$I(H) = \int_0^H (1 + x(t)^2)^{-5/2}\, dt. \tag{10}$$

Here C/H is a residence fraction, while I/H is the mean host density relative to its central value. Neither is an encounter probability. The comparisons for nonzero kicks are

$$I_{\text{central}}(H) = H, \tag{11}$$

$$I_{\text{wait}}(H) = \max(H - P, 0); \tag{12}$$

both equal H for a stationary object.

The production calculation uses energy quadrature with absolute and relative tolerances of 10^{-11}. Exposure integrals are split at radii 0.01, 0.1, 1, 10, and 100 when these lie on the orbit. This resolves narrow central passages at large apocenter. Complete periods accumulate exactly; the partial period is recovered by a bracketed inversion of outgoing travel time. Core crossing times are found by quadrature, not by a time-sampled indicator.

TABLE I. Representative first-return results. The apocenter is in units of b; P is in units of t_d.

η	x_a	P
0.5	0.881917	4.395561
0.8	2.591534	11.371439
0.9	5.167285	27.972769
0.95	10.207544	74.156668
0.99	50.241305	792.551744
0.999	500.249126	24856.344701

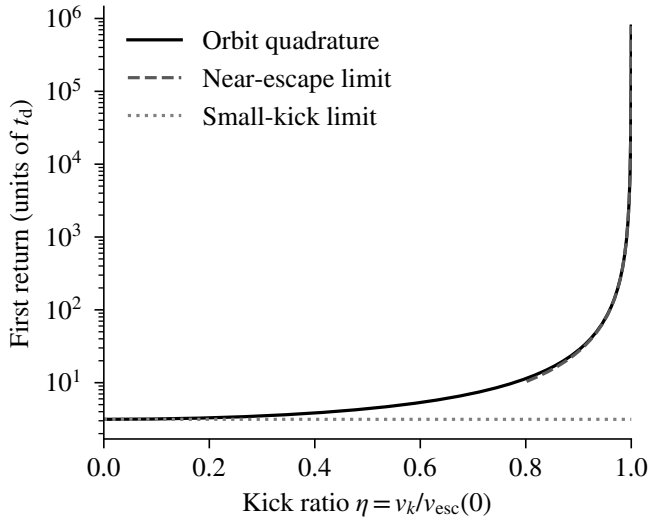


FIG. 1. First central-return time from energy quadrature, with small-kick and near-escape limits. Binding alone does not imply a short excursion. The return is a crossing, not dissipative settling.

III. FIRST-RETURN RESULTS

The return benchmark tabulates nine kick fractions from 0.1 to 0.999. The plotted curve extends to 0.9999. Table I lists representative results, and Fig. 1 compares the numerical curve with its asymptotic limits. All lengths and times use the units in Sec. II.

The small-amplitude equation is harmonic, so $P \to \pi$ as $\eta \to 0^+$. At exactly zero kick, the particle instead remains stationary. Near escape,

$$P \sim \frac{\pi}{\sqrt{2}}(1 - \eta^2)^{-3/2}. \tag{13}$$

TABLE II. Density exposure and residence over one radial period. All entries are dimensionless; $C(P,1)/P$ is the fraction of a period inside $r < b$.

η	$I(P)$	$I(P)/P$	$C(P,1)/P$
0.3	2.774830	0.793277	1
0.8	1.369228	0.120409	0.174465
0.99	1.029215	0.00129861	0.0019312

At $\eta = 0.99$, the apocenter is about $50b$ and the excursion takes about $793t_d$. Such large excursions also expose the limitations of the isolated potential: tides, external structure, or host evolution could materially alter a real remnant's trajectory. The tabulated values are conditional on the stated ideal model.

IV. RESIDENCE AND EXPOSURE RESULTS

The exposure comparison uses 11 kicks from 0 to 0.999 and spheres with $c = 0.1, 0.3, 1$. Seven fixed horizons span 1 to 1000 dynamical times. For the ten nonstationary kicks, a second grid uses $H/P = 0.1, 0.5, 1, 2, 10$. Fixed and period-scaled windows are distinct comparisons.

At $\eta = 0.99$, the time inside $r < b$ is 1.530574 dynamical times per radial period. The mean sampled density is approximately 0.130% of the central value, while the residence fraction inside this sphere is 0.193% (Table II). These percentages describe one ideal orbit, not an observed population.

At $H = P$, the wait prescription gives zero, whereas the orbit has accumulated $I(P) > 0$. At $H = 10P$ for $\eta = 0.99$, the orbit gives 10.292151 dynamical times of exposure and the wait prescription gives 7132.965700. The error changes sign (Fig. 2). More generally, as H grows, the wait prescription approaches unit mean density while the orbit approaches $I(P)/P < 1$ for every nonzero kick. It is therefore not uniformly conservative.

An orbit stays within a sphere when its apocenter does not exceed its radius:

$$\eta \le \sqrt{1 - \frac{1}{\sqrt{1+c^2}}}. \tag{14}$$

For $c = 1$, the threshold is 0.541196. Equality touches the sphere boundary instantaneously and does not change an integrated residence fraction. Even an orbit entirely within the sphere samples a varying density; residence and density exposure are different quantities. Figure 3 shows the dependence on the operational sphere radius.

V. VELOCITY-WEIGHTED COMPANION ACCESS

A. Specified population and comparison

We set $M = 10^6\,M_\odot$, $b = 1\,\mathrm{pc}$, and remnant mass $m_\mathrm{r} = 60\,M_\odot$. An imposed population of $N_\mathrm{b} = 100$ tracer binary black holes has total mass $m_\mathrm{b} = 60\,M_\odot$ per binary and semimajor axis $a = 0.1$ AU. Its mass is 0.6% of the fixed host mass. The unsegregated target number density is

$$n_\mathrm{b}(r) = n_0\widetilde{\rho}(x), \qquad n_0 = \frac{3N_\mathrm{b}}{4\pi b^3}, \quad \widetilde{\rho}(x) = (1+x^2)^{-5/2}. \tag{15}$$

The target center-of-mass velocities follow the isotropic equilibrium Plummer distribution, not a measured black-hole subsystem.

All four cases, $\eta = 0, 0.3, 0.8, 0.9$, share $H = 10$, corresponding to $T = Ht_d = 0.149096\,\mathrm{Myr}$. They are admitted only if their apocenters lie within five Plummer half-mass radii, $6.52383b$. This is an operational restriction on the benchmark domain, not a physical tidal boundary; neither the potential nor the target distribution is truncated. The $\eta = 0.99$ exposure example in Sec. IV fails that screen and remains an isolated-model limit. A common elapsed time is essential: the higher-kick cases have not completed one return by $H = 10$, whereas the $\eta = 0.3$ case has completed about 2.86 radial periods.

A close passage is defined by two-body monopole pericenter $R = 1$ AU from the target binary center of mass. This is ten binary semimajor axes, a boundary around the binary's neighborhood rather than a cross section for exchange. For relative speed w before mutual focusing,

$$\Sigma(w) = \pi R^2\left[1 + \frac{2G(m_\mathrm{r}+m_\mathrm{b})}{Rw^2}\right]. \tag{16}$$

The local two-body approximation neglects host tides during the approach and internal binary multipoles. It does not determine the outcome after entry.

B. Velocity weighting and the density approximation

Let u be target speed and v remnant speed in the host frame. At fixed radius, $y = u/v_\mathrm{esc}(r)$ has density

$$p(y) = \frac{512}{7\pi}y^2(1-y^2)^{7/2}, \qquad 0 \le y \le 1. \tag{17}$$

This follows from the isotropic Plummer distribution $f(E) \propto (-E)^{7/2}$ [9]. Isotropic angular averaging gives

$$\langle w\rangle_\Omega = \max(v,u) + \frac{\min(v,u)^2}{3\max(v,u)},$$
$$\langle w^{-1}\rangle_\Omega = \frac{1}{\max(v,u)}. \tag{18}$$

The velocity-integrated zero-speed limit is finite. We evaluate the remaining speed moments with incomplete beta functions and check them by direct numerical angle and speed integration.

Define the local encounter kernel and passage rate by

$$K(r,v) = \pi R^2\left[\langle w\rangle + \frac{2G(m_\mathrm{r}+m_\mathrm{b})}{R}\langle w^{-1}\rangle\right],$$
$$\Gamma(r,v) = n_\mathrm{b}(r)K(r,v),$$
$$\tau(T) = \int_0^T \Gamma[r(t_\mathrm{phys}), v(t_\mathrm{phys})]\,dt_\mathrm{phys}. \tag{19}$$

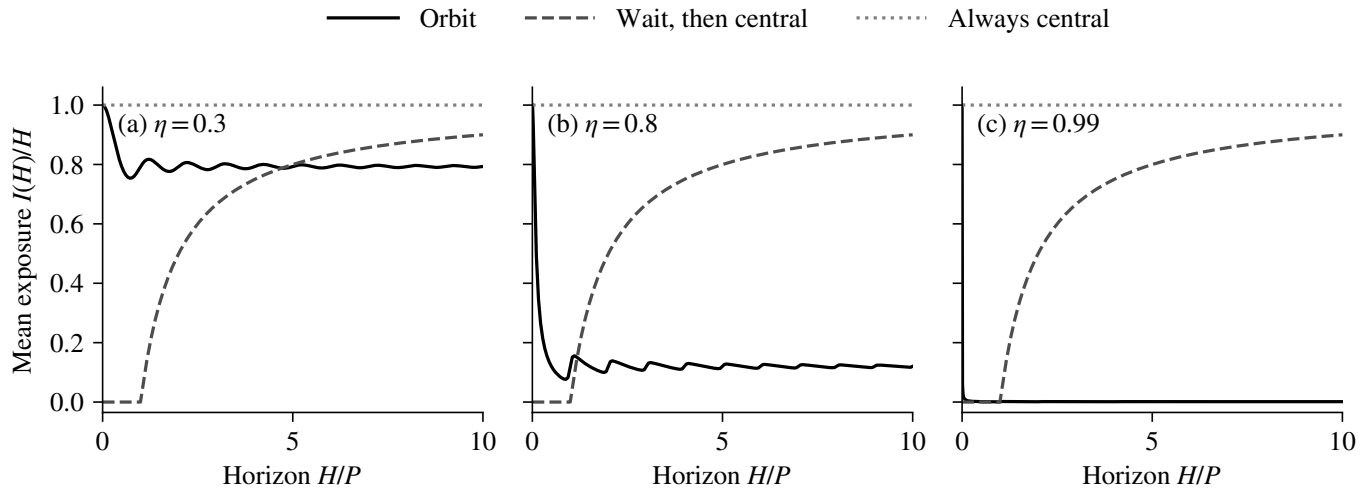


FIG. 2. Mean normalized density sampled up to each horizon for three kick fractions: (a) $\eta = 0.3$, (b) $\eta = 0.8$, and (c) $\eta = 0.99$. The first-return comparator suppresses all initial exposure and then assumes unit density permanently. The undamped orbit repeatedly departs from the center. Line styles distinguish the prescriptions independently of color.

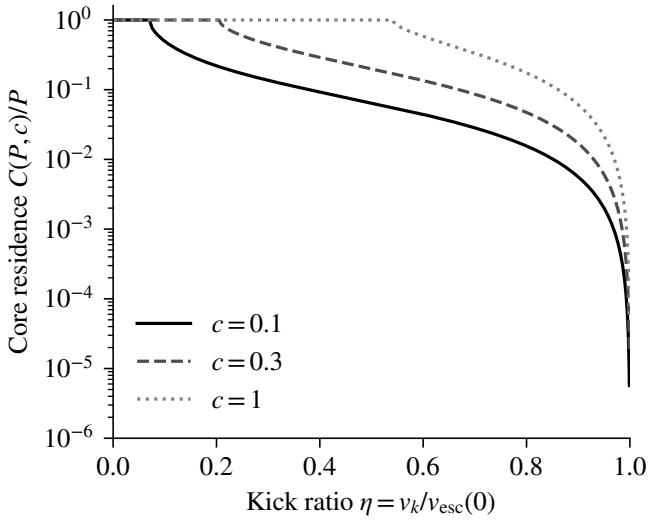


FIG. 3. Residence fraction per radial period for three operational sphere radii. Each curve equals unity below the threshold in Eq. (14) and decreases as bound kicks approach escape. The radii are model definitions, not measured cluster core radii.

Here K has dimensions of volume per time, Γ inverse time, and τ is a dimensionless expected passage count along the unperturbed orbit. We do not turn τ into a merger probability or assume that the orbit remains unchanged after an actual encounter.

For implementation, use $V = \sqrt{GM/b}$ and dimensionless $t = t_{\rm phys}/t_d$. The geometric and focusing integrals are

$$J_g = \int_0^H \widetilde{\rho}\langle w/V\rangle\, dt, \qquad J_f = \int_0^H \widetilde{\rho}\langle V/w\rangle\, dt,$$
$$\tau = n_0 \pi R^2 b \left(J_g + \lambda J_f\right), \qquad \lambda = \frac{2(m_{\rm r} + m_{\rm b})b}{MR}. \tag{20}$$

The length factor $b = Vt_d$ applies the time conversion once. In this configuration $\lambda = 49.5036$.

The stationary central control defines $Q = \tau(\eta, H)/\tau(0, H)$. The density proxy is $D = I(H)/H$. Their relationship is exact for the specified bath:

$$\frac{Q}{D} = \frac{\int_0^H \widetilde{\rho}[x(t)] K[r(t), v(t)]\, dt}{K(0,0) \int_0^H \widetilde{\rho}[x(t)]\, dt}. \tag{21}$$

Thus the density approximation discards variation in the encounter kernel. It is accurate when the density-weighted mean of K is close to its stationary central value; residence alone does not enforce that condition.

C. Results and physical interpretation

The signed proxy error is $\delta = D/Q - 1$. Before execution, we declared $|\delta| > 0.10$ a failure of this benchmark's approximation screen. This is an operational tolerance, not a significance level or a requirement from gravitational-wave inference.

Table III and Fig. 4 show errors of +4.41%, +53.06%, and +98.55% at the three nonzero kicks. The two larger kicks fail the screen. At $\eta = 0.9$, density exposure is 5.81% of the control but passage count is 2.93%. The density-only approximation is nearly twice the conditional velocity-weighted count. The ordering across kicks does not reverse.

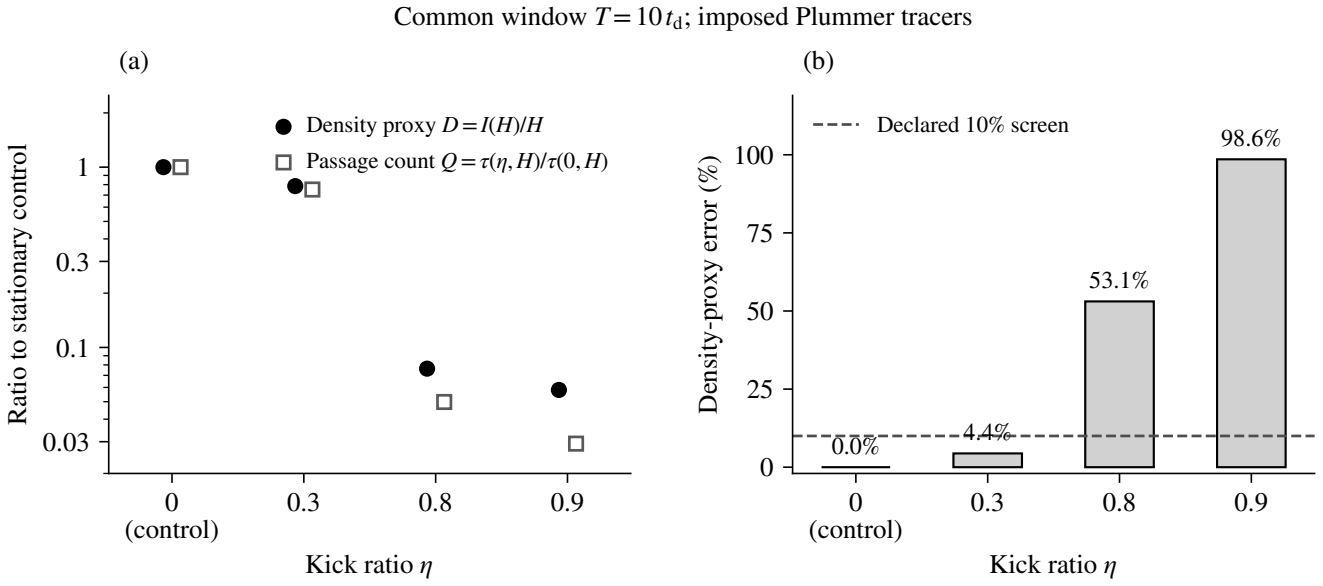


FIG. 4. Velocity-weighted access: (a) normalized density exposure D and passage count Q over the same elapsed time; (b) signed density-proxy error and the predeclared 10% screen. The central stationary case is the normalization control. Values are deterministic calculations, without sampling error bars.

TABLE III. Companion-access comparison at the common horizon $H = 10$. D and Q are normalized to the stationary control; $\delta = D/Q - 1$ is reported in percent. τ is an expected passage count, not an observed count or merger probability.

η	D	Q	δ (%)	τ
0	1.00000	1.00000	0.00	1.608×10^{-6}
0.3	0.78472	0.75156	4.41	1.209×10^{-6}
0.8	0.07626	0.04982	53.06	8.011×10^{-8}
0.9	0.05811	0.02927	98.55	4.706×10^{-8}

The geometric and focusing terms respond differently to motion. At $\eta = 0.9$, their separately normalized contributions are 0.1110 and 0.02867, respectively, compared with $D = 0.05811$. Higher relative speed increases geometric flux but reduces the inverse-speed contribution of focusing. Focusing supplies 97.23–99.19% of the nonzero-kick counts and controls the total discrepancy. Slow motion near apocenter does not compensate for the low tracer density there.

Absolute counts remain small: the stationary control has $\tau = 1.6080 \times 10^{-6}$ and the $\eta = 0.9$ case has $\tau = 4.7064 \times 10^{-8}$. These are conditional outputs for the imposed population and window, not an encounter-rate prediction for a calibrated cluster. A large fractional proxy error here does not establish a large merger-rate effect. The short window also does not prove that friction, diffusion, or host evolution would be negligible in a real system.

VI. CLUSTER POPULATION AND POTENTIAL COMPARISON

A. Reference state and comparison design

We use the full evolved example distributed with `cmctoolkit` [10]: model `N8-RV1-RG8-Z0.1`, snapshot `initial.snap0382.dat.gz`, at the pinned source revision identified in Ref. [8]. It was selected as an available evolved example before examining discrepancies. It is one simulated state, not an observational dataset or the full CMC catalog [11].

The age is 12.7323 Gyr, total system mass $M = 224633\,M_\odot$, and three-dimensional half-mass radius $r_h = 4.79771$ pc. Matching these last two quantities gives Plummer scale $b = r_h\sqrt{2^{2/3} - 1} = 3.67706$ pc. This comparison uses current properties, not initial conditions or the imposed bath's host parameters. The numerical kick fractions recur across the studies, but their physical speeds are not the same across hosts.

We separately count all systems, all binaries, single black holes, binaries containing at least one black hole, and binaries containing two black holes (BBHs). Stellar-type code 14 identifies black holes in this input convention. The declared minimum count of 30 screens whether continuous BBH-distribution calibration is supported; it is not a statistical sufficiency theorem. Descriptive radial and speed CDF distances carry no independent-sample significance interpretation.

TABLE IV. Evolved-snapshot inventory. Radius is the number-median system radius divided by the host's mass-weighted half-mass radius. Categories overlap: BBHs are a subset of BH-bearing binaries.

Population	N	median$(r)/r_h$
All systems	682,583	1.24677
All binaries	31,122	1.09910
Single BHs	21	0.04018
BH-bearing binaries	6	0.03851
BBHs	2	0.01014

B. What the companion sample supports

There are 682583 systems, 31122 binaries, and 29 black-hole objects. Only two systems are BBHs (Table IV). Their total masses are 26.672 and 29.821 $M_\odot$, and their semimajor axes span 0.23958–12.40947 AU. Their median radius is 0.04864 pc, or $0.01014 r_h$. All binaries and all black holes would be inappropriate replacements for this specific target population.

These two centrally concentrated binaries do not calibrate the fixed-mass, fixed-separation, unsegregated bath of Sec. V. No continuous distribution is fitted. This snapshot also cannot reject such parameters for every cluster or age, or establish a universal BBH distribution. Concentration is consistent with established mass-segregation physics [4], not a newly identified mechanism. Speed fractions in the recorded audit use the matched Plummer escape speed; their discrepancy mixes host-potential and population assumptions and cannot alone diagnose velocity anisotropy or equipartition.

C. Frozen spherical potential and identical kicks

Each system is spread over a spherical shell at its snapshot radius. With zero potential at infinity,

$$\Phi_{\rm sh}(r) = -G\left[\frac{M(<r)}{r} + \sum_{r_i \geq r} \frac{m_i}{r_i}\right]. \tag{22}$$

The central value uses the finite limiting shell sum. Exterior mass contributes to potential energy even where its force vanishes. For central launch speed v_k, the energy is $\mathcal{E} = v_k^2/2 + \Phi_{\rm sh}(0)$; the apocenter solves $\Phi_{\rm sh}(r_a) = \mathcal{E}$, and

$$P_{\rm sh} = 2\int_0^{r_a} \frac{dr}{\sqrt{2[\mathcal{E} - \Phi_{\rm sh}(r)]}}. \tag{23}$$

We split the integral at shell boundaries and regularize the turning point. This is a frozen monopole calculation with no drag or encounters.

The snapshot and matched Plummer central escape speeds are 26.547 and 22.924 km s^{-1}. We apply identical physical kicks, chosen as 0.3, 0.8, and 0.9 times the latter speed. Renormalizing by each model's own escape speed would change the physical experiment. All three kicks are bound in both isolated potentials. This binding classification is distinct from tidal removal in CMC [7].

TABLE V. First ballistic returns at identical physical kicks within each model pair. $P_{\rm sh}$ uses 2048 compressed shells; $P_{\rm Pl}$ is the matched Plummer value. Times are Myr and speeds are km s^{-1}.

$\eta_{\rm Pl}$	v_k	$P_{\rm sh}$	$P_{\rm Pl}$	$P_{\rm Pl}/P_{\rm sh}$
0.3	6.8771	0.26250	0.77588	2.9557
0.8	18.3388	0.89636	2.52230	2.8139
0.9	20.6312	1.39333	6.20464	4.4531

Matching M and r_h does not preserve the central depth or return times (Table V, Fig. 5). Plummer return times are 2.9557, 2.8139, and 4.4531 times the shell values. All exceed the predeclared 10% return-time screen, measured relative to the shell result. This tests transfer of a potential-shape approximation to one state; it does not measure the error of a live-cluster trajectory model.

CMC models relaxation and strong encounters but samples new orbital phases during its Monte Carlo evolution [7]. Connecting snapshot radii is therefore not a continuous physical recoil trajectory. The frozen comparison neither follows a displaced remnant in a live host nor evaluates its encounters with the measured targets. Those missing calculations prevent an astrophysical calibration of the imposed-bath passage counts.

VII. LATE-TIME AVAILABILITY OF BINARY TARGETS

A. Objective and sample definition

The full snapshot leaves a population question that a denser kick grid cannot answer: is its small BBH target count confined to one saved state? We audit every standard-model summary in the same pinned `cmctoolkit` release [8, 10]. Selection rules were committed before reading the additional count outcomes. The input inventory contains 7537 summaries across 126 model histories; six separately tuned NGC 6624 models are excluded. The published CMC catalog contains 148 simulations [11], but not all appear in this late-time summary release. Unavailable histories are not assigned zero populations, and their absence does not establish disruption.

The primary comparison selects one saved state per model: the state closest to 12 Gyr within an inclusive 0.25 Gyr window, resolving ties by earlier age and then source path. This avoids weighting models by their output cadence. Seven histories have no eligible state; the 119 selections span 11.8013–12.2086 Gyr. Equal model weights define a descriptive old-cluster comparison, not a probability distribution for observed clusters or merger epochs. Integer populations are not interpolated between saved states.

The target count is the number of binaries whose two stellar types are both 14. Total black-hole objects, BH–non-BH binaries, and all binaries remain separate populations. We check type-count sums, binary-pair accounting, the identity between

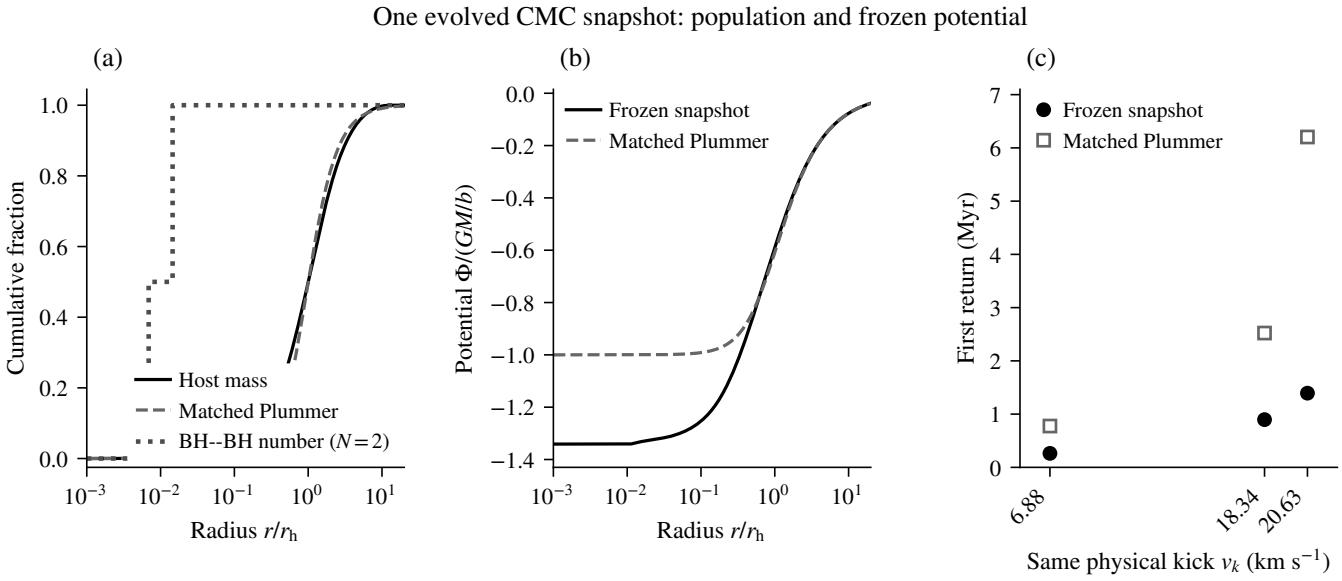


FIG. 5. Snapshot comparison: (a) host mass fractions and the number fraction of the two BBHs; (b) frozen snapshot and matched Plummer potentials; (c) first returns at identical physical kicks. The BBH curve describes two systems, not a fitted companion distribution. Potentials and returns use the 2048-shell representation.

TABLE VI. Late-time target-availability audit. The primary comparison uses one state per model near 12 Gyr. Missing age coverage is separate from a measured zero count. The count-30 screen is operational, not a sufficiency theorem.

Quantity	Count
Available model histories	126
Validated saved states	7,537
Selected near 12 Gyr	119
Histories without age coverage	7
Selected $N_{\rm BBH}$: min/median/max	0/2/9
Selected states with $N_{\rm BBH} = 0$	38
Selected states with $N_{\rm BBH} \leq 2$	71
Selected states with $N_{\rm BBH} \geq 30$	0
Maximum $N_{\rm BBH}$ over saved states	15

system and object counts, and nonnegative single-BH membership. Every source path and age is retained. Model-level minima, medians, and maxima describe available saved states, not time averages or counts of distinct objects. Without binary identities, repeated appearances cannot be treated as new companions.

B. Results and scope of the sparsity finding

The selected states contain between zero and nine BBHs, with median two (Table VI, Fig. 6). There are 38 measured zero states and 71 states with at most two BBHs. By contrast, total BH-object counts range from zero to 1878 with median 38, and all-binary counts range from two to 142232 with median 25867. Neither larger population substitutes for the specified BBH targets. All 119 selected states fall below the inherited count-30 screen.

The complete set of saved summaries spans 10.0007–14.0022 Gyr and has maximum BBH count 15; no inspected state reaches the screen. These are simulation output ages, not a cosmological-age selection. The model used in Sec. VI has 26 saved states with one to three BBHs and a saved-state median of two. Its near-12 Gyr state contains three BBHs at 11.9449 Gyr; its original full snapshot contains two at 12.7323 Gyr. The small count therefore extends beyond the initially inspected state and throughout the available late-time release. This does not establish universal sparsity at other ages or in nuclear clusters.

The summaries provide aggregate counts but not the joint positions, velocities, binary masses, semimajor axes, eccentricities, or stable identities required to calibrate the passage calculation. None passes the operational single-state count screen; more fundamentally, abundance alone cannot supply those missing correlations. Sparse individual states may still admit a useful smooth ensemble description. A measured zero means no instantaneous target of this class, not that future formation or other acquisition channels are absent. Earlier cluster calculations already found small numbers of coexisting BBHs and rapid population turnover [5]; the present contribution is the reproducible scope and accounting of this pinned release. This census itself does not determine target lifetimes, turnover, or the probability of encounter during a remnant's return.

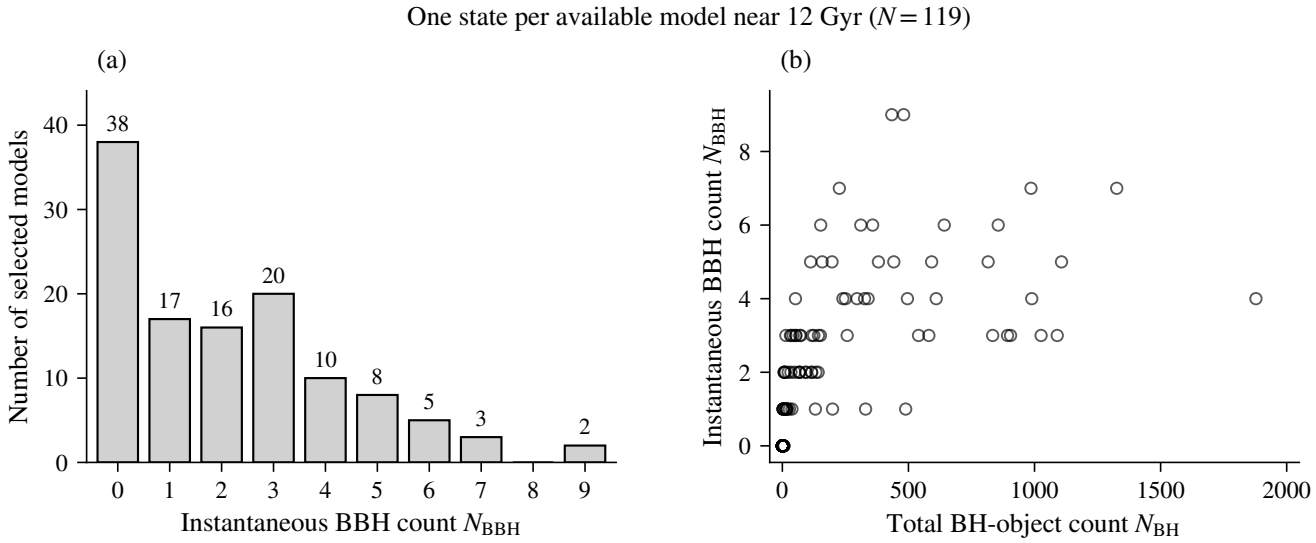


FIG. 6. Binary-target availability in 119 age-selected model states: (a) integer BBH-count distribution; (b) BBH systems versus all BH objects in the same states. Coincident points overlap; neither panel fits a population law. The operational count-30 screen lies beyond the displayed BBH range. These model-grid counts have no independent-observation confidence intervals.

VIII. ANALYTICAL AND NUMERICAL VERIFICATION

For a stationary object, $I(H) = C(H, c) = H$. For nonzero bound kicks, both exposure and occupancy repeat every P, hence $I(nP) = nI(P)$. In the small-kick limit, $I(P) \to \pi$ and $I(P)/P \to 1$.

Near escape, the marginally bound outgoing trajectory obeys $\dot{x} = \sqrt{2}(1 + x^2)^{-1/4}$. The limiting exposure of a full bound excursion is

$$\begin{aligned}\lim_{\eta\to 1^-} I(P) &= \sqrt{2}\int_0^\infty (1+x^2)^{-9/4}dx \\ &= \sqrt{\frac{\pi}{2}}\frac{\Gamma(7/4)}{\Gamma(9/4)} \\ &\simeq 1.0166557018. \end{aligned} \tag{24}$$

Combining this with Eq. (13) gives

$$\frac{I(P)}{P} \sim \frac{\Gamma(7/4)}{\sqrt{\pi}\Gamma(9/4)}(1-\eta^2)^{3/2}. \tag{25}$$

This is a limit from bound orbits. An exactly marginally bound particle does not return in finite time.

The complete verification suite contains 180 passing tests. Twenty-seven establish the orbital benchmarks. Independent checks integrate the signed equations of motion using DOP853 [12], compare central crossings, monitor energy, and integrate accumulated density as an additional state variable. Core residence is checked with event-segmented trajectories. Other tests cover stationary behavior, sphere thresholds, period accumulation, crossing continuity, density as the derivative of exposure, invalid inputs, and convergence near escape. The strongest near-escape numerical cases reach $\eta = 0.9999$; no claim is made of uniform numerical accuracy arbitrarily close to escape.

Thirty-five checks of the passage calculation include independent numerical velocity integrals, analytical stationary and fast-motion limits, and separately integrated DOP853 trajectories with numerical velocity moments. Agreement at 10^{-8} relative tolerance satisfies the predeclared 10^{-6} requirement. Other checks cover units, population/radius/mass scaling, period accounting, and the apocenter screen. Production orbit integrals use absolute and relative tolerances of 10^{-10}.

Forty-six tests check snapshot interpretation and shell calculations against direct potential sums, force derivatives, analytical one-shell returns, independent adaptive quadrature, smooth-Plummer recovery, scaling, and tied empirical-CDF jumps. Eight derived snapshot quantities agree with the authors' separately tabulated summary: age, total mass, black-hole mass, system count, object count, binary count, black-hole count, and BBH count. Counts match exactly; relative tolerances are 10^{-7} for masses and 10^{-12} for age. This validates interpretation against another processing of the same state, not an independent astrophysical dataset. Mass columns are already physical; velocities use the N-body time conversion and ages the distinct relaxation-time conversion. External conversion files are parsed without execution.

Approximately equal-mass shell bins use mass-weighted harmonic radii, preserving total mass and central potential. We compare 256, 512, 1024, and 2048 shells. The final two resolutions differ by less than 0.005% in return time and 0.023% in apocenter. Adjacent-resolution differences need not bound error. A direct comparison with all 680110 unique shell radii gives return-time errors of 0.1555%, 0.00131%, and 0.000568%, respectively, below the 1% numerical screen and far below the potential-shape discrepancies. The finite-

shell innermost cavity is force-free; its arbitrarily small-kick limit is not the smooth-core harmonic limit.

The catalog audit adds 67 synthetic accounting and selection tests and five archive-integrity tests. Every consumed file matches its locked Git object and recorded SHA-256 checksum. All 7537 states pass count accounting. A separately implemented scalar extraction and age-selection calculation reproduces the primary census. Seven quantities also agree with the independently parsed full snapshot: age, host mass, system count, all binaries, single BHs, BH-bearing binaries, and BBHs. That comparison reuses the same physical state and is not an additional cluster sample.

Stored-result verification reproduces 146 orbital/exposure rows, four passage cases, five snapshot cohort summaries, twelve compressed-shell cases, 160 profile points, and the full-shell checks. It also reproduces all catalog rows, 126 history summaries, and 119 selected states. Reproduction with the same implementation is distinct from the independent checks above. Grid points, resolution levels, and repeated snapshots are not independent astrophysical observations; solver tolerances are not physical-model uncertainties.

IX. DISCUSSION AND CONCLUSIONS

A. Claims supported by the combined evidence

The orbital benchmarks show that a first-return-then-central prescription does not preserve exposure: it suppresses early sampling and eventually overestimates late sampling. The near-escape return time diverges while exposure per bound excursion remains finite. These are properties of the collisionless benchmark, not frictional settling estimates.

The passage calculation identifies another omission. Equation (21) shows that density weighting discards changes in the relative-velocity kernel. The selected larger kicks fail the operational approximation screen even when the orbit is accurately computed. This establishes a conditional approximation error, not a universal direction or magnitude of astrophysical bias.

The cluster comparisons constrain inputs that numerical accuracy alone cannot justify. Matching host mass and half-mass radius fails to preserve the selected snapshot's ballistic returns. Its two BBH targets cannot calibrate the imposed bath. The late-time census broadens the small-count evidence to the inspected release, without identifying the joint companion distribution. Changing the prescribed number of targets to the census median would leave velocities, concentration, binary properties, and merger conditioning unresolved. Likewise, multiplying the passage errors by the frozen-potential return ratios is unjustified: the comparisons use different hosts and do not calculate their coupled dynamics.

Together, the results identify three distinct requirements for a reduced accessibility model: adequate remnant motion, velocity-dependent passage weighting, and a target population appropriate to the remnant's environment and epoch. This is the supported synthesis. It does not establish a measured reacquisition rate, an observational formation-channel discriminator, or a new mechanism for hierarchical mergers.

B. A hypothesis for merger-conditioned accessibility

The next theoretical question is whether those ingredients can be reduced independently. Let $\mathcal{M}$ specify a preceding merger, its host state and remnant properties, conditional on retention. Let $N_{\rm b}(t)$ be the number of available BBH targets at elapsed time t, excluding the consumed progenitor binary and any system containing the tagged remnant. For a specified local passage model, define

$$\begin{aligned} p_n(t \mid \mathcal{M}) &= \Pr[N_{\rm b}(t) = n \mid \mathcal{M}], \\ g_n(t \mid \mathcal{M}) &= \mathbb{E}[\Gamma(t) \mid N_{\rm b}(t) = n, \mathcal{M}]. \end{aligned} \tag{26}$$

The ensemble mean predicted passage count over a common physical window T is then

$$\overline{\tau}(T \mid \mathcal{M}) = \int_0^T \sum_{n=0}^{\infty} p_n(t \mid \mathcal{M}) g_n(t \mid \mathcal{M})\, dt. \tag{27}$$

The rate g_n incorporates target phase space and binary properties together with the remnant path. Equation (27) is a conditional-expectation identity, not an additional simulation result or an assumption of Poisson encounters. Zero targets imply zero instantaneous passage rate for this channel, but subsequent formation can change the population. A mean count alone does not determine the probability of at least one encounter.

Our working hypothesis is that, at fixed host and remnant properties, the temporal overlap of remnant motion with changing target availability provides predictive information about finite-window passage counts beyond a stationary target description and first-return time. The physical motivation is that a returning remnant samples companions present along its path during the relevant interval; an age-averaged population need not preserve that overlap. The present calculations establish neither its sign nor its size. In particular, the census measures equal-weight late-time states, not $p_n(t \mid \mathcal{M})$, and no live trajectory is supplied for g_n.

This is a proposed test of a reduced description, not a claim that coupled population dynamics are absent from existing cluster models [5, 7, 11]. An earlier assembly model already permits new hard binaries to form while a retained remnant is inactive during recentering [1]. Successive-merger and evolving-cluster frameworks also incorporate encounter and population physics [2, 3]. The intended advance is a quantified domain of validity for reducing that physics to remnant accessibility. That advance would require validation beyond the present benchmarks.

C. A comparison that could reject the hypothesis

The required reference sample consists of merger-linked cluster histories with stable binary identities and adequately resolved target formation, exchange, destruction and ejection. Full snapshots must retain joint center-of-mass position and velocity, component masses, semimajor axis and eccentricity. Both younger and older epochs may contribute; selection at a fixed late age cannot substitute for conditioning on a

remnant-producing merger. Parent binaries and consumed objects must be excluded from the subsequent target inventory, and repeated appearances must retain their identities.

First isolate population modeling: supply the same independently resolved remnant paths and the same close-passage definition to a stationary-bath model and a model retaining target-count evolution and joint properties. Match the host, remnant properties, observation window, and marginal target abundance, so a difference cannot be attributed solely to changing the normalization. Hold out entire cluster histories when assessing predicted counts against recorded encounters. Count-only corrections and corrections to joint phase space should also be assessed separately. Improvement must exceed numerical and ensemble uncertainty and persist on held-out histories, rather than arise only from additional fitting freedom.

The hypothesis would be disfavored in a declared regime if the stationary description already predicts passage counts within the required accuracy and a time-dependent description yields no stable predictive improvement, with sufficient precision to resolve a predeclared effect size. Conversely, a reproducible residual tied to target availability during displacement, removed by the extended model, would support retaining that information. Accuracy requirements, comparison statistics and sample selection must be fixed before those runs. The very small imposed-bath counts illustrate why a few zero-encounter paths would be uninformative; adequate ensemble size and a justified count model are essential. No such held-out comparison has been executed here.

Then isolate the trajectory approximation with the population treatment held fixed. Live collisional integrations, paired kicked and un-kicked runs, independent microstates, and varied kick directions can compare frozen-potential and central-residence prescriptions with inward core crossing after exit, residence, and recorded passages. Escape must be separated from finite-window non-return. Integration, close-encounter regularization, and changes to relaxation under particle-number scaling require independent checks. Phase-resampled CMC radii cannot supply continuous recoil paths [7].

The broader objective is a defensible connection between assembly environments and repeated-merger source populations. The present result is an evidence-based narrowing of that problem to companion accessibility. Even a successful accessibility model would still require exchange or capture, binary evolution, and merger outcomes before supporting a gravitational-wave population prediction.

DATA AND CODE AVAILABILITY

The code, configurations, numerical results, and manuscript source supporting this work are available in the public black-hole-recoil-dynamics repository.

Python 3.11 is the numerical reference environment, with pinned dependencies. Study manifests record generating revisions, source/output hashes, and runtime versions. The cluster audits also pin the external simulation revision and every consumed file's checksum. The paper manifest binds all study inputs to generated tables, vector figures, the PDF, and the source archive. Project-authored code and documentation use the MIT license. Upstream data and dependencies retain their own rights; raw external files are fetched into verified local cache rather than redistributed. No observational data are used.

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